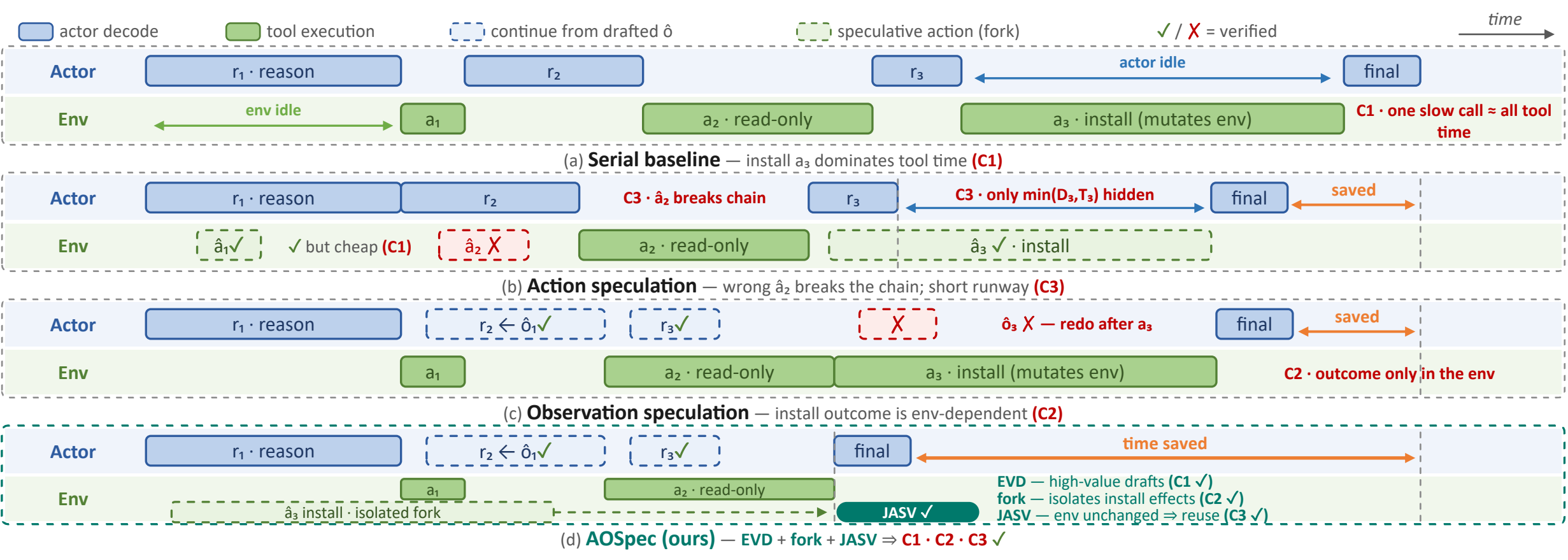

actor decode
tool execution
continue from drafted ô
speculative action (fork)
✓ / ✗ = verified
time
Actor
Env
r₁ · reason
r₂
r₃
actor idle
final
env idle
a₁
a₂ · read-only
a₃ · install (mutates env)
C1 · one slow call ≈ all tool time
(a) Serial baseline — install a₃ dominates tool time (C1)
C3 · â₂ breaks chain
C3 · only min(D₃,T₃) hidden
saved
â₁✓
✓ but cheap (C1)
â₂ ✗
â₃ ✓ · install
(b) Action speculation — wrong â₂ breaks the chain; short runway (C3)
r₂ ← ô₁✓
r₃✓
ô₃ ✗ — redo after a₃
C2 · outcome only in the env
(c) Observation speculation — install outcome is env-dependent (C2)
time saved
â₃ install · isolated fork
JASV ✓
EVD — high-value drafts (C1 ✓)
fork — isolates install effects (C2 ✓)
JASV — env unchanged ⇒ reuse (C3 ✓)
(d) AOSpec (ours) — EVD + fork + JASV ⇒ C1 · C2 · C3 ✓